

A complex network diagram with nodes and edges, overlaid with logos and text. The nodes are arranged in a hierarchical, tree-like structure, with many nodes containing binary strings (e.g., 0123456789, 01234567, 0123456, 012345, 01234, 0123, 012, 01, 0). The edges are colored in various shades of purple, blue, and green. The diagram is centered on the slide, with the title text overlaid on it.

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A data structure for monomial ideals with applications to signature Gröbner basis

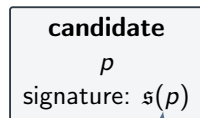
Pierre Lairez, Rafael Mohr & Théo Ternier
MATHEXP, Inria Saclay, France

ICMS 2026

Waterloo, Canada – July 21, 2026

Motivation: processing of signatures

Inside the main loop in sGB algorithm (Faugère, 2002)

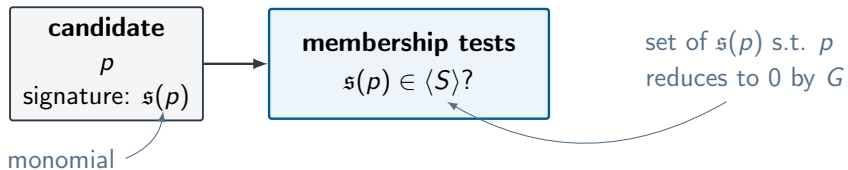


monomial



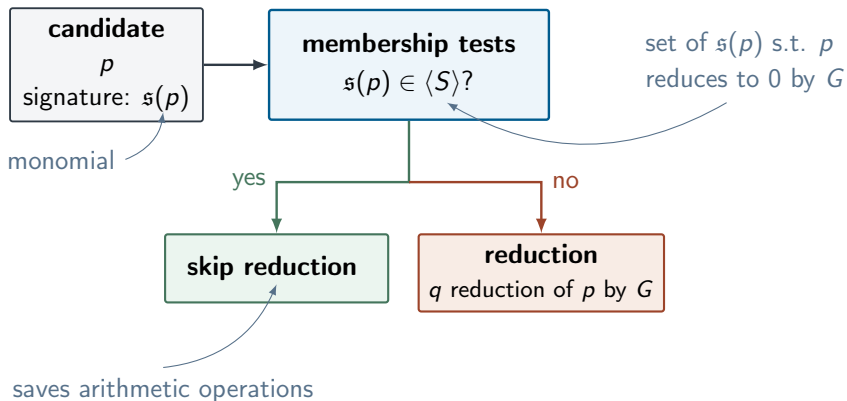
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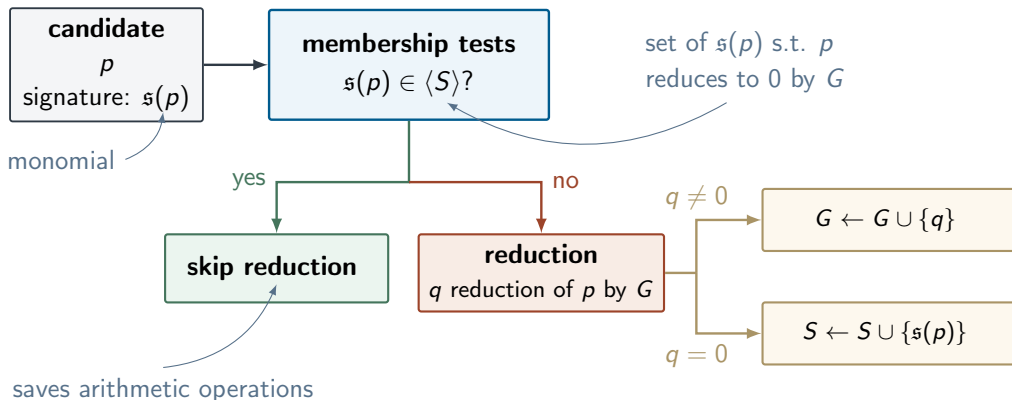
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Inside the main loop in sGB algorithm (Faugère, 2002)

candida

p

signature:

monomial

eco¹⁴: 14 polynomials in 14 variables

All reductions to 0 avoided but...

117,861,802

membership tests in monomial ideals

(*AlgebraicSolving.jl*)

$G \cup \{q\}$

$\cup \{s(p)\}$

saves arithmetic operations

Membership test

Goal: Efficiently test monomial membership in a monomial ideal.

Example: $I = \langle x^2z^3, x^2y^3z, x^3, yz^2 \rangle$ $m = x^2y^2z^2 \in I?$

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If n number of variables and N number of generators:

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Our data structure Independent of N

If d is the largest generator degree:

$$O(n \log d)$$

Practical impact

eco¹⁴ runs **3× faster**

- **Divmasks:** Prune some divisibility checks.

For each monomial a store a 64-bit integer $\text{div}(a)$ such that for any monomial b ,

$$a \mid b \implies (\text{div}(a) \& \sim \text{div}(b)) == 0$$

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Dubé, 1993

Construction: $O(N \log(N)^{n-1})$

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- **BDD-engine** for computations on monomial ideals.

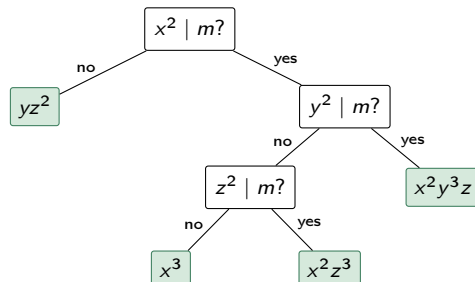
Moreno-Resa, Sáenz de Cabezón, ISSAC 2026

- Kd-trees

Roune, Stillman, ISSAC 2012

kd-tree of $I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle$

$m = x^2y^2z^2?$



Membership: $O(nN)$

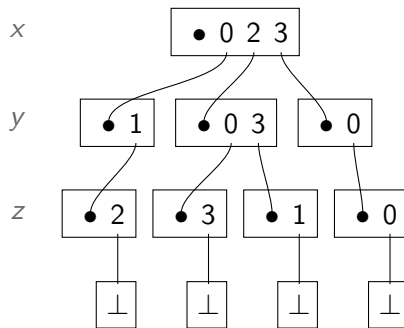
State of the art

- Prefix tree

Milowski, 2004

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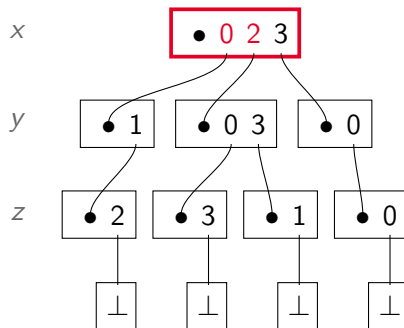
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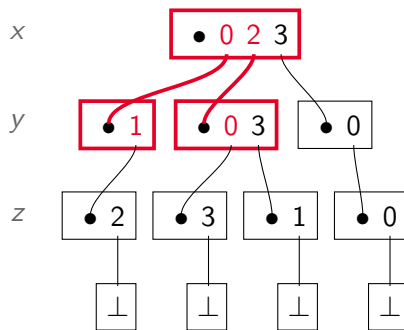
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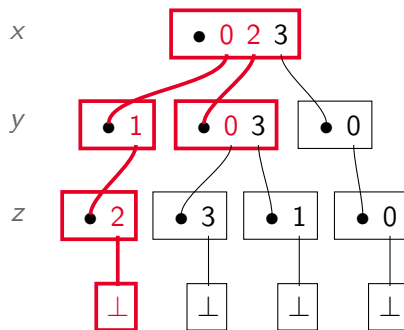
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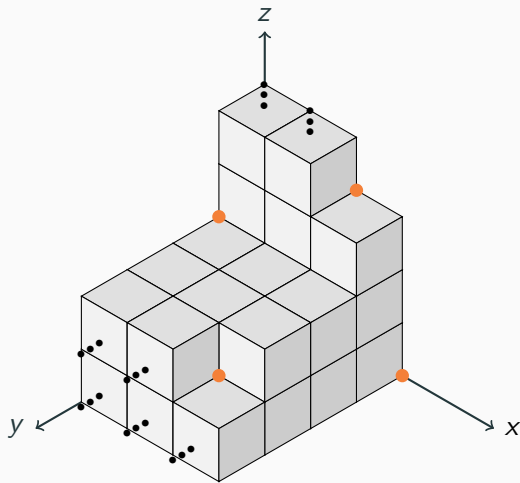


Construction: $O(nN)$

Membership: $O(nN)$

Geometric representation: staircase of $I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle$

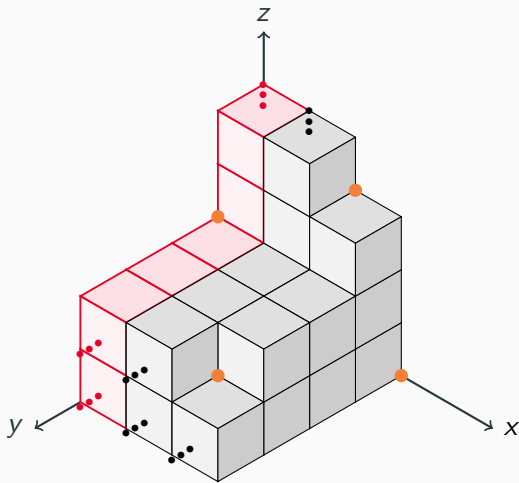
$$I/k = \langle m \in \text{Mon}(y, z) \mid x^k m \in I \rangle$$



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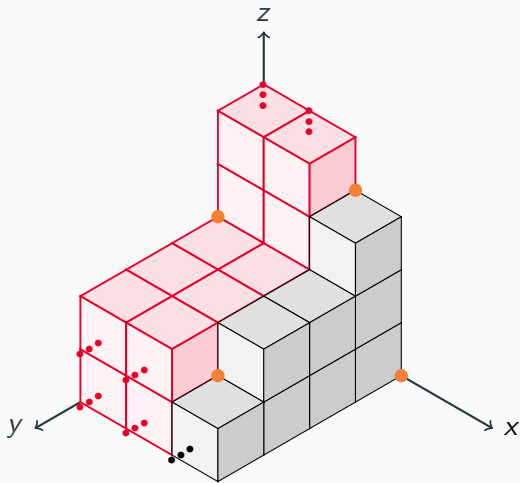
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$\cap$

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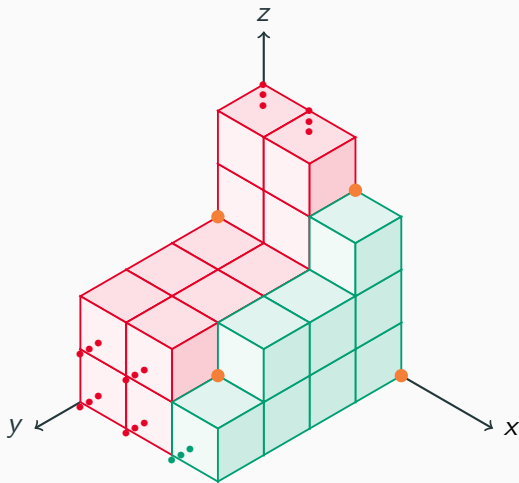
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$$I/2 = \langle yz^2, z^3, y^3z \rangle$$



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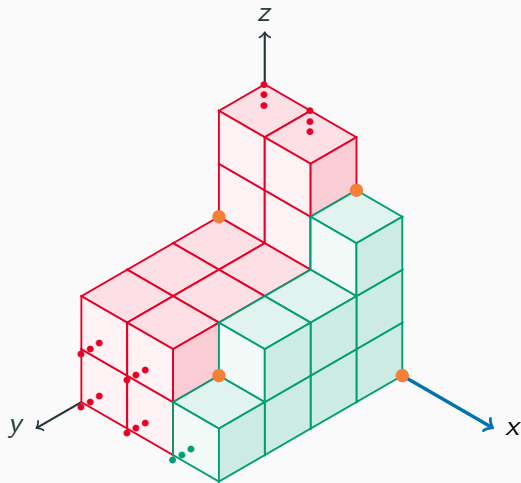
$$I/1 = \langle yz^2 \rangle$$

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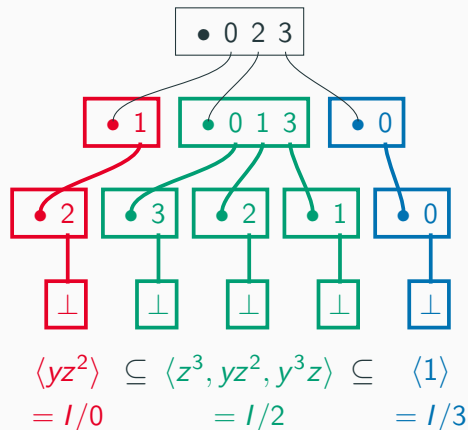
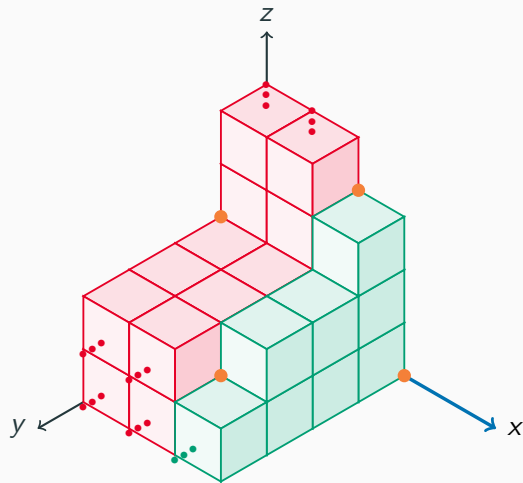
$$I/2 = \langle yz^2, z^3, y^3z \rangle$$

$\cap$

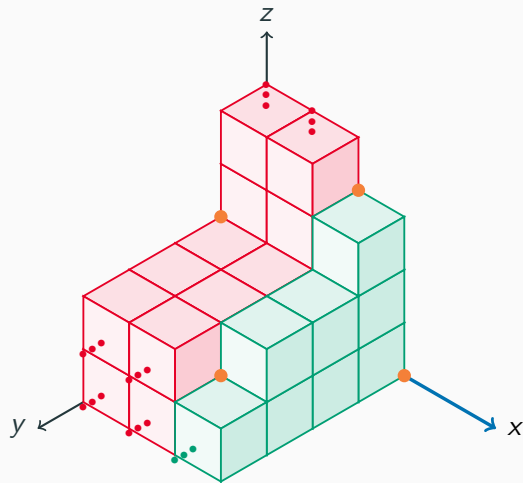
$$I/3 = \langle yz^2, z^3, y^3z, 1 \rangle$$



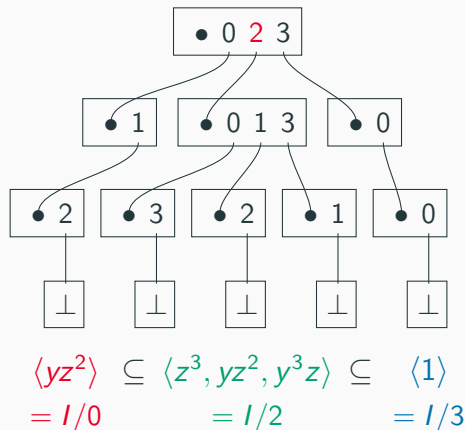
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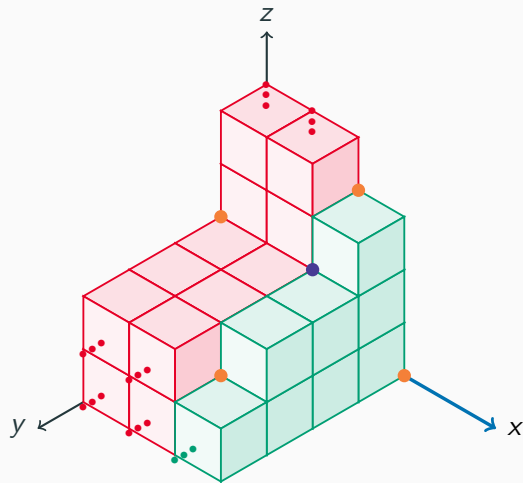
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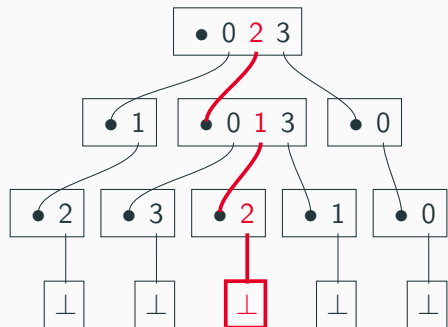
Membership: $m = \mathbf{x}^2 y^2 z^2 \in I?$



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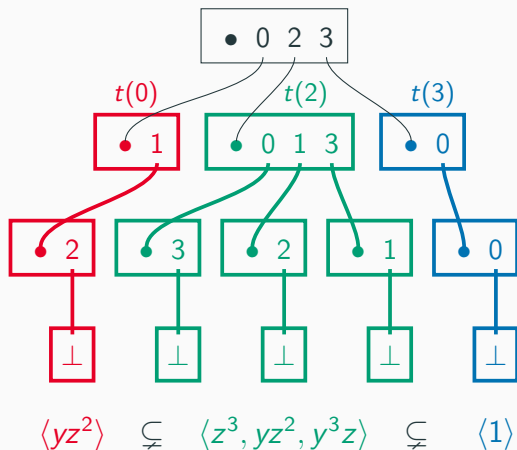


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$$\begin{aligned} \langle yz^2 \rangle &\subseteq \langle z^3, yz^2, y^3z \rangle \subseteq \langle 1 \rangle \\ &= I/0 &= I/2 &= I/3 \end{aligned}$$

Ideal n -trees

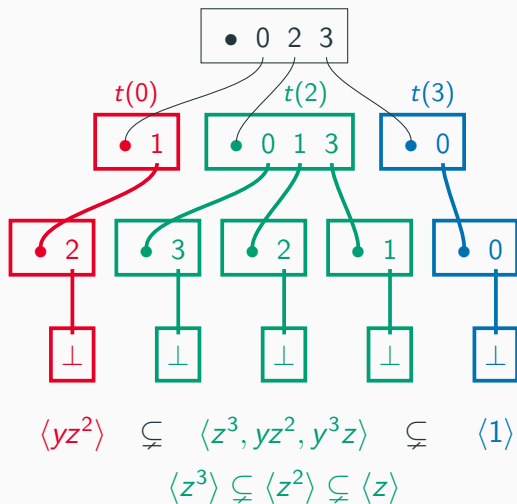


Ideal n -trees

A n -tree t is ideal if $n = 0$ or $n > 0$ and

- $\forall e, f \in \text{dom}(t),$
 $e < f \implies \mathbb{I}(t(e)) \subsetneq \mathbb{I}(t(f))$

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- $\forall e \in \text{dom}(t),$
 $t(e)$ is ideal

Insertion to build an ideal n -tree

In sGB computations: monomials are inserted one by one

Let $m = x_1^{m_1} \dots x_n^{m_n}$

Ideal n -tree of $\langle m \rangle$:



Insertion to build an ideal n -tree

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I a monomial ideal, t its ideal n -tree. We define:

$\text{ins}(t, m)$

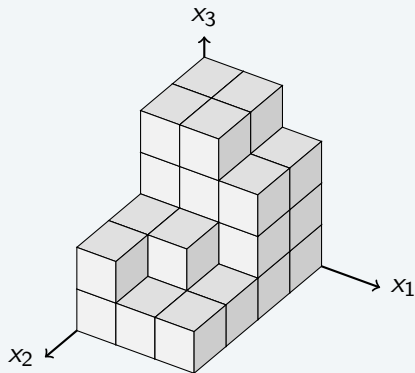
the ideal n -tree of $I + \langle m \rangle$

Ideal n -tree of $\langle m \rangle$:



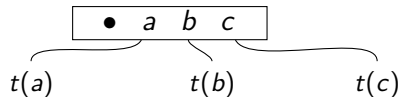
Insertion of $m = x_1^{m_1} m'$ in the ideal n -tree t of I

Geometric intuition



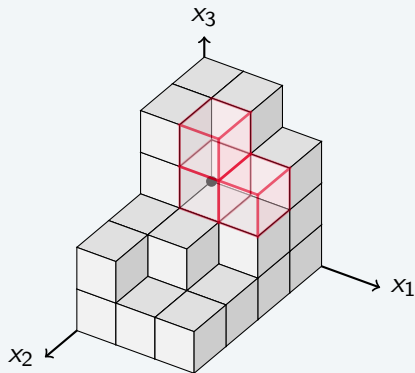
before insertion

Tree before insertion



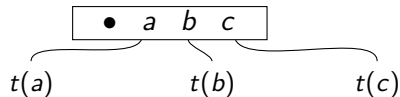
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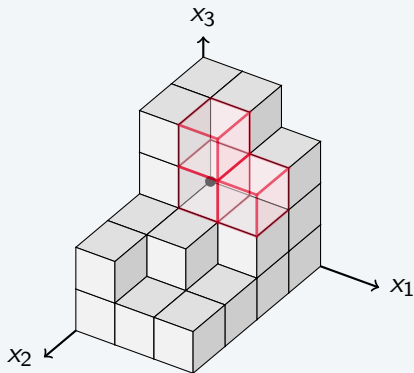
insert $m = x_1^{m_1} m'$

Tree before insertion



Insertion of $m = x_1^{m_1} m'$ in the ideal n -tree t of l

Geometric intuition



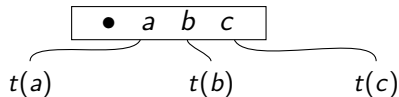
update slices $e \geq m_1$

Quotients

$$e < m_1 : (l + \langle m \rangle) / e = l / e$$

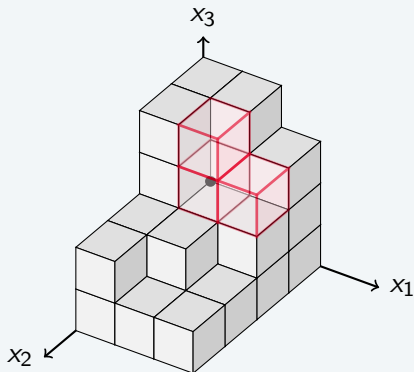
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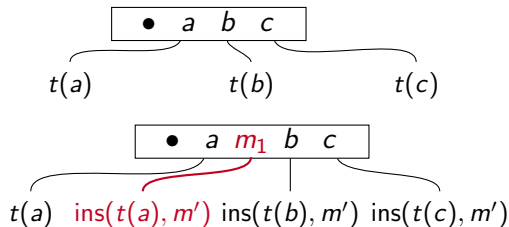
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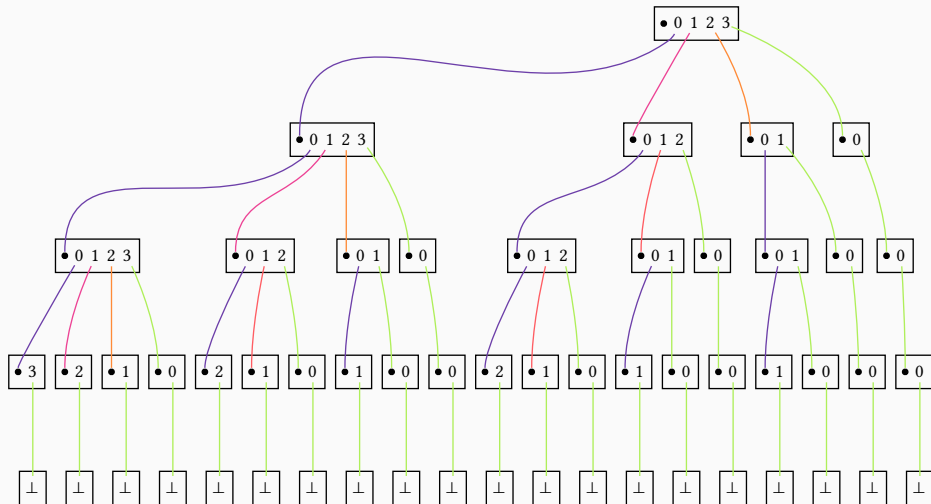
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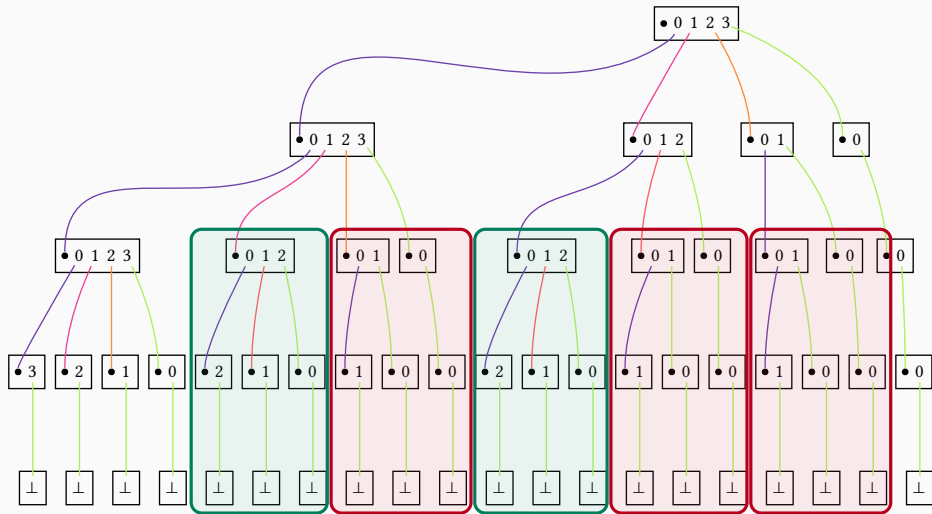
Tree update after insertion



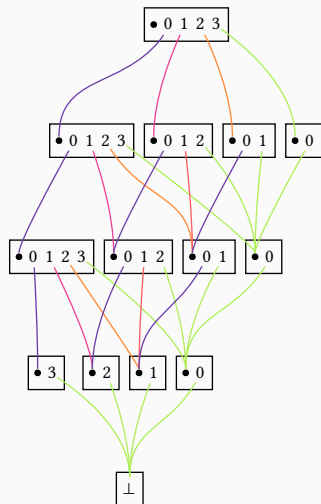
Problem of width for ideal n -trees



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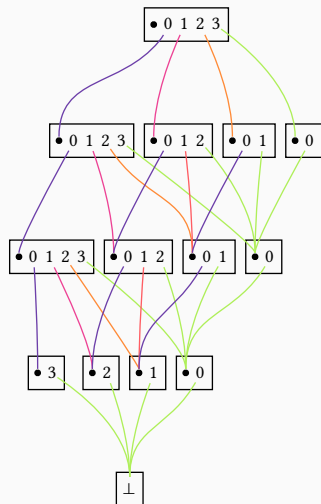
Solved problem of width with maximal sharing



Maximal sharing

Invariant: *structurally equal iff physically equal.*

Solved problem of width with maximal sharing

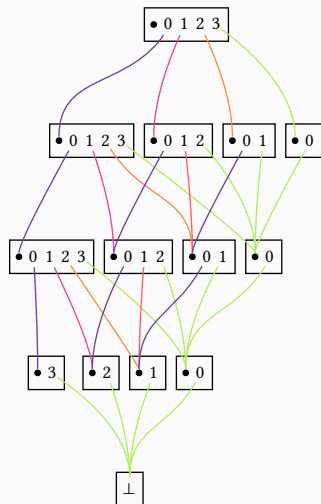


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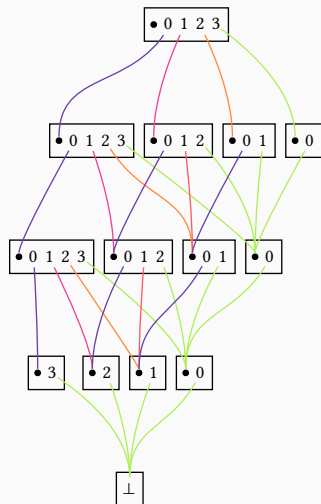


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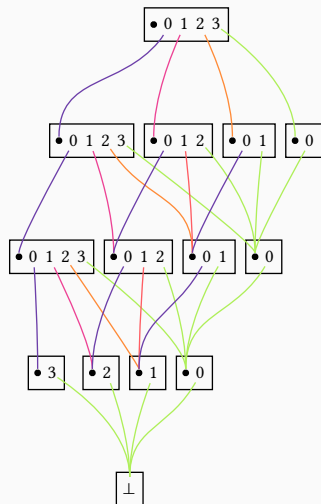


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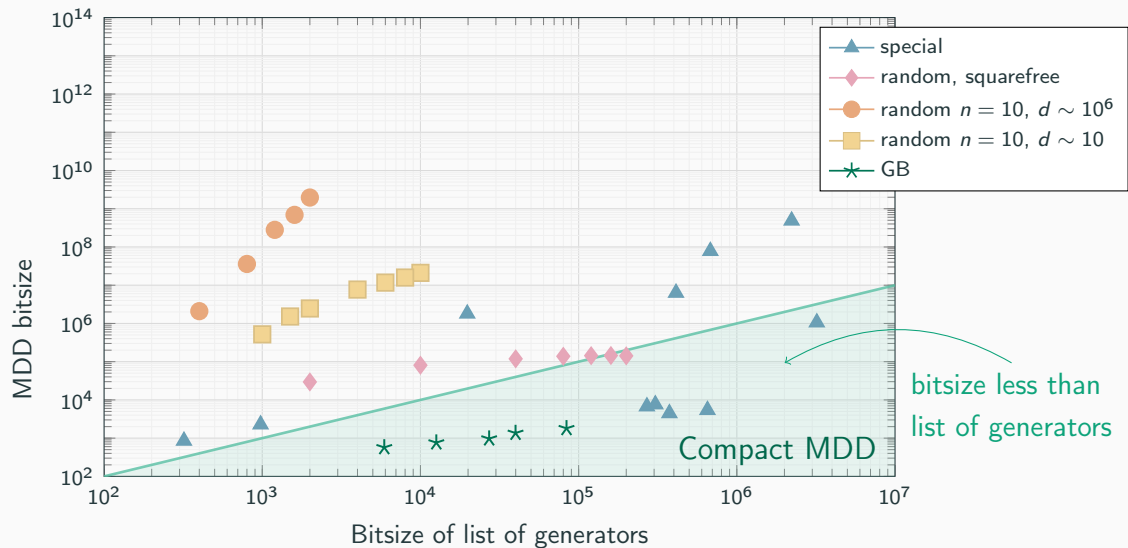
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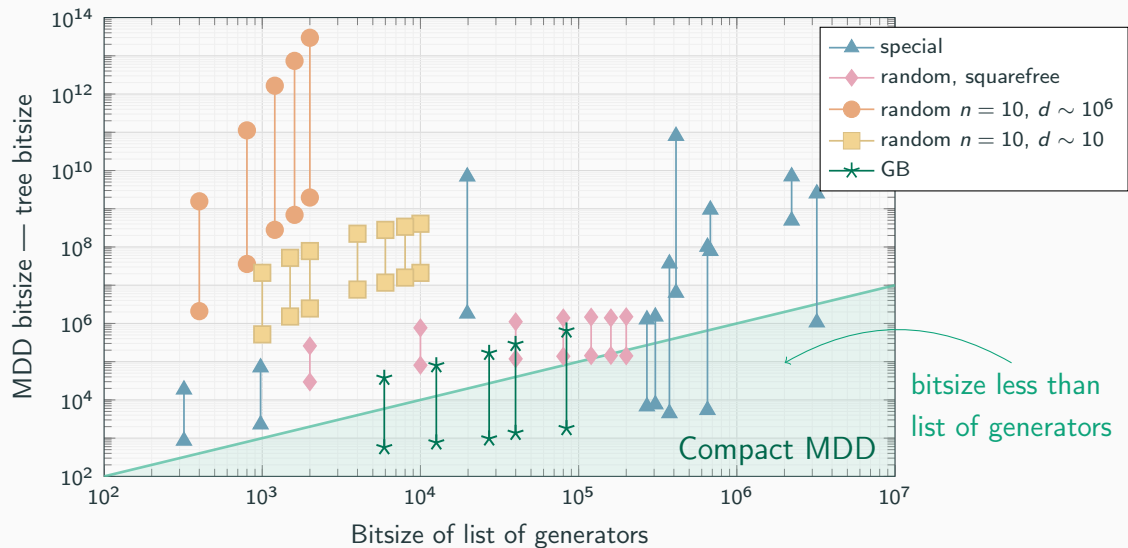
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Result: Manipulation of ideal n -trees much faster and with less memory.

Experimental data on the bitsize of MDDs



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Initial ideal of eco^{14} (from Gröbner basis computations)



Bitsize of list of generators: 39928

Bitsize of MDD: 1360

Benchmarks on *AlgebraicSolving.jl*

name	<i>AlgebraicSolving.jl</i>		<i>AlgebraicSolving.jl</i> with MDD						
	time	% sig.	time	% sig.	test	MDD	size	nb tests	×
katsura ¹¹	4s	86%	2s	79%	0.04s	0.06s	81	388k	1.8
katsura ¹²	27s	83%	14s	70%	0.17s	0.27s	103	1.5M	2.0
katsura ¹³	3m21	82%	1m47	69%	0.71s	1.14s	125	5.9M	1.9
cyclic ⁷	1s	98%	0.8s	98%	0.02s	0.02s	43	240k	1.6
cyclic ⁸	1m08	95%	32s	89%	0.50s	0.36s	80	4.1M	2.1
eco ¹⁴	3h22	90%	1h09	68%	5.79s	0.92s	173	118M	2.9
noon ⁷	10s	87%	3s	60%	0.08s	0.02s	71	985k	3.1
noon ⁸	2m34	82%	44s	34%	0.63s	0.10s	101	7.4M	3.5
noon ⁹	1m18	99%	1m23	99%	0.58s	0.10s	71	7.2M	0.9

AlgebraicSolving.jl

Christian Eder (Kaiserslautern), Mohab Safey El Din (Sorbonne Université), Rafael Mohr (Leuven) and Rémi Prébet (Lyon).

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MDD bitsize

Initial ideals from GB computations:

- Generic coordinates:
Borel-fixed (Galligo, 1974)

$$O(ndN)$$

- Generic systems: conjectured to be
Weakly revlex (Moreno–Socias, 1991)

$$O(nd^2)$$

Find reducer

Go beyond yes/no membership:

$m \in I \rightsquigarrow$ minimal generator index

Improve the symbolic part

Applications in commutative algebra

Hilbert series